The Statistics of Travel Time

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1 Introduction

When taking public transport, in previous years, it had been historically unpredictable. Some days, nothing would go wrong and it would take 40 minutes. On worse days, it could take over an hour and a half. This led me to having an issue with arriving at school on time and varies significantly with some days being 20 minutes late to school. My junior year I changed from taking public transportation to taking an electric scooter. This greatly reduced the time to 20 minutes and was more reliable. As a result, I was on time to school significantly more often and if I were late, it was usually within 5 minutes of school starting. My experiences with public transport are what inspired me to figure out the distribution of me arriving at school and how often I should end up on time.

Statistics can give very interesting insights into many things as many things in the world are uncertain. A key concept of statistics necessary to calculate how often I will end up at school on time are probability density functions and the center of probability density functions is a normal distribution. A normal distribution is one of the most commonly used probability density functions. Probability density functions, PDFs, give probabilities to models which are not discrete.

Within statistics, there are two types of probability functions. The two types are discrete and continuous. Many probabilistic models are discrete, such as a roll of the dice or a coin flip. These have definite results and there is a countable number of outcomes, however this can't apply to most things in life. To model the time it takes to travel from A to B, this can't be The travel time from A to B, the position of electrons, and the time of decay for a single radioactive atom are all events that can't be represented by a discrete probabilistic model as they can be measured infinitesimally. This means that any event for any specific time would have a 0% chance off happening, yet it occurs, so we know that there is a chance. With probability density functions finding the area between two values you get the probability that it falls in between those two numbers. But if you take a single point then you get a zero probability, fixing this conundrum.

The daily trip I take consists of 4 parts: one bus, two trains, and a walk. For starters I take a bus to Harvard Square. From there I take a train, called the Red Line, to Park Street. Then I switch to another train called the Green Line, which I take to Copley. From there I walk the rest of the way to school, which is about a 10 minute walk. For the sake of simplicity, my variation will be ignored, which means that there will be no error for walking and I will always be on time for the bus. This paper has a focus on the variation due to public transportation, rather than from me.

Solving this problem with many interconnected distributions is a difficult analytical problem. It is analytically highly complex. A Monte Carlo simulation enables one to estimate these results accurately with much less complexity and with greater ease.

2 Normal Distribution

A normal distribution is one of the most used probability density distributions used. A normal distribution is used for measurement errors to modeling the distribution of test scores for SAT. The normal distribution is the basis for a significant part of statistics, and can be found in practically any area of study where statistics are used. The derivation of the equation is a long and a little too complex so, it will have to be taken for granted and the equation and graph will be shown below.

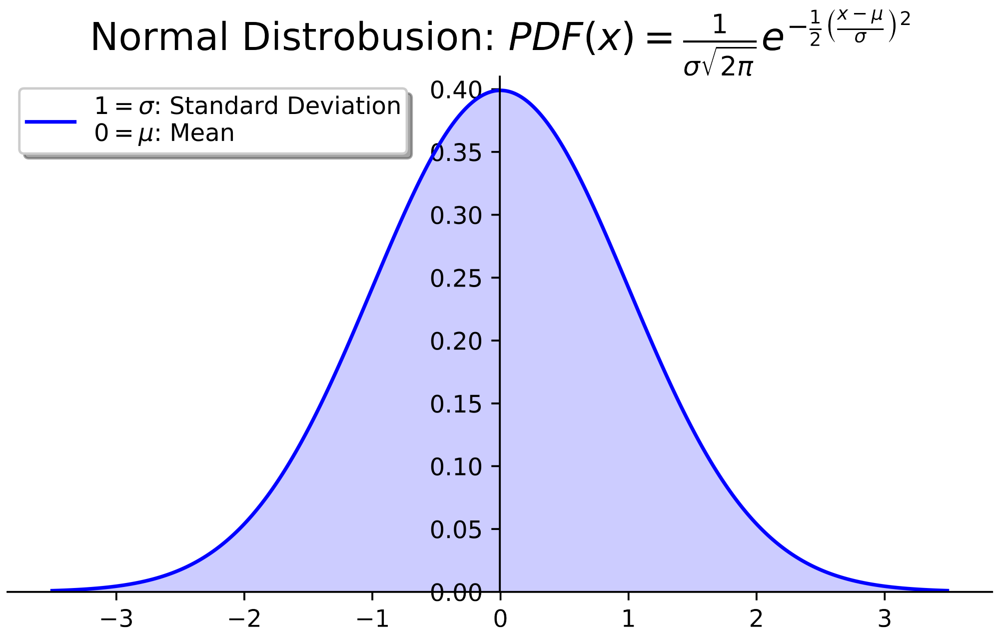

Figure 1

The normal distribution is important for many reasons. It is a great method for showing variation and representing a data set. For the purposes of modeling the transportation, a normal distribution is going to resemble the arrival time of a train or bus. The calculations for fitting the data to the model is below (1). However as figure 2 shows, a normal distribution will not be capable of modeling the data for travel time.

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$
$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\mu - x_i)^2} \tag{1}$$

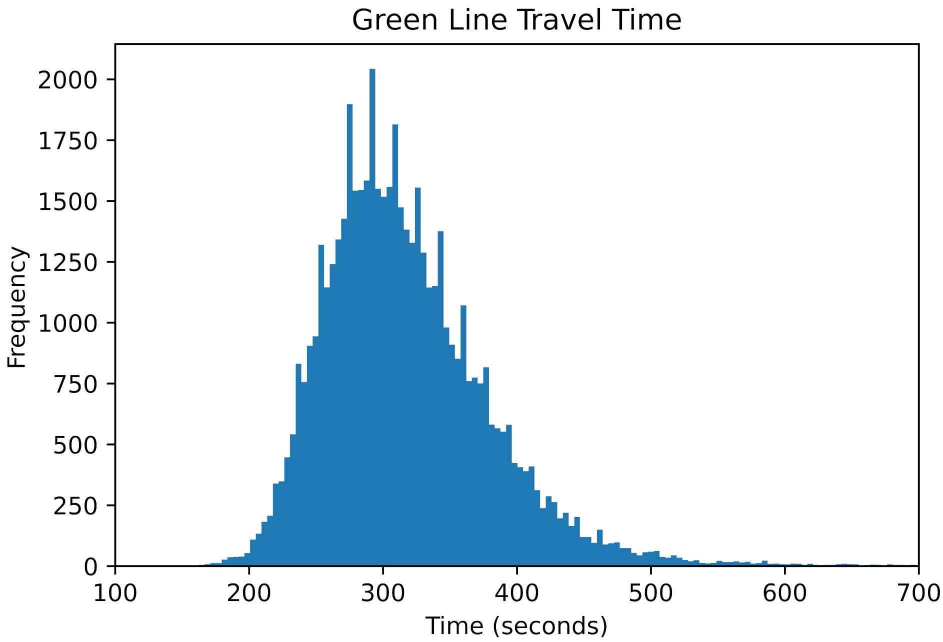

Figure 2

The data for figure 2, is the travel time of a specific train taken, which is called the Green Line. With any sort of transportation there is a minimum travel time, which a normal distribution could not accomplish. A normal distribution can propagate scenarios of negative travel time, which is an impossibility. Furthermore, as can be seen in figure 2, there is a preference for the right than the left side of the peak. Both of these factors cause it to resemble a log-normal distribution. A later example of a log-normal distribution, figure 3, can be used as a comparison.

3 Log-Normal Distribution

The log-normal distribution is derived from the normal distribution. The main characteristic of the log-normal distribution that makes it unique to a normal distribution is that the number is greater than zero. So when dealing with things in the real world, where we don't have negative values. Since it has to be greater than zero, the log-normal distribution is ideal for modeling anything that has a value that can't be negative.

The way it is derived is quite easy from the normal distribution[1].

$$\begin{gathered} f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} \\ \\ x \in \mathbb{R} \\ f_X(x) > 0 \end{gathered} \tag{2}$$

Since the desired characteristics is that it will only output for values greater than zero, then we need to find a function to input as x such that it takes the function outputs the set of real numbers ($\mathbb{R}$), but takes in only values within the set of real numbers that are greater than zero.

$$\begin{aligned} x &\in \mathbb{R} \\ g(x) &= e^x \\ g(x) &> 0 \end{aligned} \tag{3}$$

$$\begin{aligned} &\{x | x \subseteq \mathbb{R}, x > 0\} \\ &g^{-1}(x) = \ln(x) \\ &g^{-1}(x) \in \mathbb{R} \end{aligned} \tag{4}$$

Now that we have a function that takes in a value greater than zero and outputs all values in the set of real numbers. However we can't input this into the normal distribution PDF, but we can input it since it may give the same values back, it would distort the area. A cumulative probability function, CDF, is the integral of PDF. Instead, using the CDF, we don't

[1]Nicholson, E. (2014, September 21). *The log-normal distribution.* Youtube. https://youtu.be/cLFcqKHT1uY.

distort the probability since its max is at 1 (100%).

$$
\begin{aligned}
F_X(x) &= \int f_X(x)dx \\
x &= g^{-1}(y) \\
F_Y(y) &= F_X(g^{-1}(y)) \\
f_Y(y) = \frac{dF_Y(y)}{dy} &= \frac{dF_X(g^{-1}(y))}{dg^{-1}(y)} \times \frac{dg^{-1}(y)}{dy} \\
\frac{dF_X(g^{-1}(y))}{dg^{-1}(y)} &= f_X(g^{-1}(y)) \\
\frac{dg^{-1}(y)}{dy} &= \frac{d}{dy}(\ln y) = \frac{1}{y} \\
f_Y(y) &= \frac{dF_X(g^{-1}(y))}{dg^{-1}(y)} \times \frac{dg^{-1}(y)}{dy} \\
&= f_X(g^{-1}(y)) \times \frac{1}{y} \\
&= f_X(\ln y) \times \frac{1}{y} \\
&= \frac{1}{y\sigma^*\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{\ln y - \mu^*}{\sigma^*})^2}
\end{aligned} \tag{5}
$$

The reason that both μ and σ both have * because they no longer can use the same value for those variables, those values need to be calculated. Figure 3 is an example of a log-normal distribution.

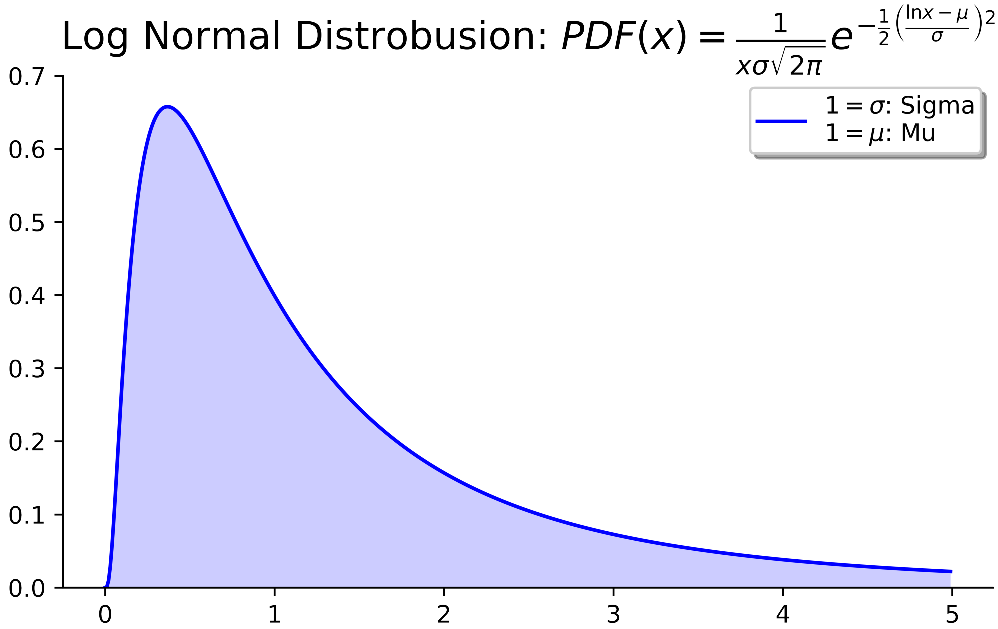

Figure 3

As the figure 3 shows, the log-normal distribution starts from 0 and goes to infinity. It shows that this succeeded at what was the goal of the function. Getting a best fit for the function is different from finding the variable for a normal distribution. However since the log-normal distribution is the essentially the natural logarithm of the normal distribution, and the standard deviation are calculated similarly to a normal distribution, however, by first taking the natural logarithm of all the data and then calculating the mean and standard deviation gives a fit for the data.

$$
\begin{aligned}
x_i^* &= \ln(x_i) \\
\mu^* &= \frac{1}{N}\sum_{i=1}^{N} x_i^* \\
\sigma^* &= \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\mu^* - x_i^*)^2}
\end{aligned} \tag{6}
$$

4 Monte Carlo

Calculating the probability of interconnected, log-normal distributed events is very challenging. A Monte Carlo is a neat method to get an accurate estimate for a solution. A Monte Carlo Simulation is a simulation that uses probabilities to simulate with what we do know. For example if one wanted to simulate the spread of light in an environment from a point source, by slightly changing the angle and simulating every position, however it would take a substantial amount of time. A much faster method would be to simulate light rays at random angles. At a small number of light rays it won't look good, however after a substantial amount of light rays, it will be quite realistic.

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The reason for the use of a Monte Carlo simulation in this internal assessment is due to the complexity of actually deriving the equation to get a proper mathematical model to determine how often I would arrive at school on time. To officially derive an equation combining multiple normal and log-normal distributions with conditions for curtain situations would take a significant amount of time and would be extremely difficult. A much simpler way to model the public transport system is to model each of the parts individually and get the overall sum of the parts in a Monte Carlo simulation. The complexity of the model is shown in figure 4.

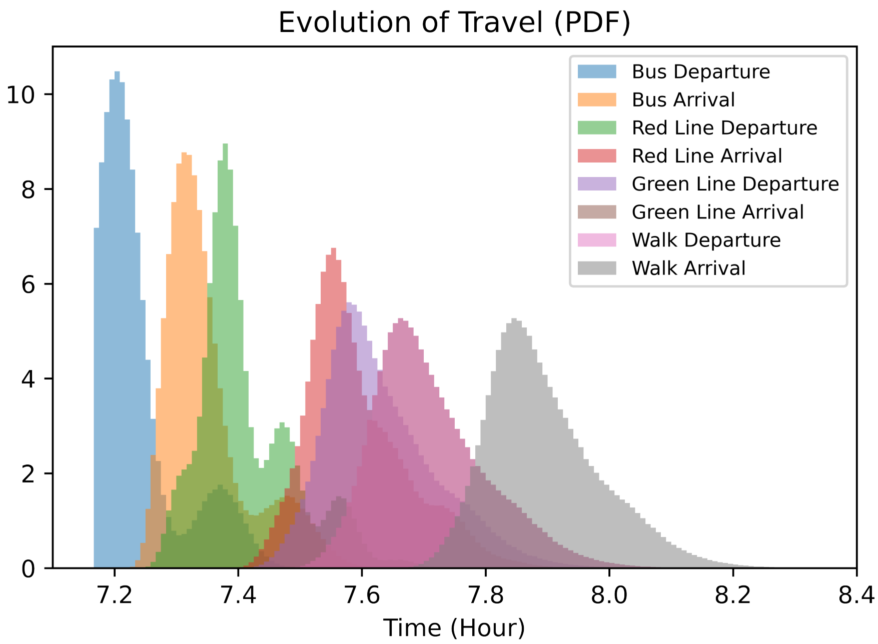

Figure 4

Figure 4 shows the evolution of the travel alternating from departure time to arrival time. The area of each histogram is one, as they are a probability density function. The model for figure 4 is explained later with great detail. As can be seen it is very complex. This model only uses normal and log-normal distributions, but is still able to show the complexity of the public transport system and the interaction from one step to the next.

In order to perform a Monte Carlo simulation, there needs to be a model for the individual parts of the system. There are two parts that need to be modeled, the time it takes for the train or bus to arrive, and the time it takes for the train or bus to get from A to B. Both of these are modeled by randomly generating a number, which resembles a distribution. If you were to generate many of these random numbers, they will approach the model. To model the time it takes to get from A to B is quite simple, because this will follow a log-normal distribution. This was shown in figure 2.

To model the waiting time is much more difficult. For this model the majority is modeled with a normal distribution, with the last being modeled with a normal distribution. The way the normal distribution was used, was that it was that a random number was generated which followed a normal distribution and if it was earlier than the arrival time, then an amount was added to the mean and this process repeated until it was later than the person arrived at the location. This works, but is not a perfect model of the wait time. It was also difficult to fit a normal distribution to the data, since there was a lot of variation in the data. The final method used to model the wait time was a normal distribution. This was calculated by calculating the time after a train arrived for there to be a train; however this is also not perfect. Either of these methods are fine as the overall model does not have to be perfect.

Now that the parts of the model have been determined, it is time to implement this using the data from the trains and buses. The data gathered is from the MBTA[2] from 2016 to 2019. 2020 is ignored in the data because when the pandemic started, this had an effect on the arrivals and departure of the trains and buses. This means it is not representative of actual results and is therefore excluded.

The exact process used was for the bus and the first train, a normal distribution was taken to determine the wait time, and the mean was increased by a specific integral and gradually tested to see if there was after the arrival of the person. The travel times were determined by shifting the data, since there is a minimum time that such an operation could possibly happen and regressing the data to a log-normal distribution. Since it is impossible to happen before this minimum, then it is shifted to reflect this so there is no chance of it being lower than this. The results are later shifted by adding the minimum to the random number generated to fix for the shift earlier.

[2]Massachusetts Bay Transportation Authority

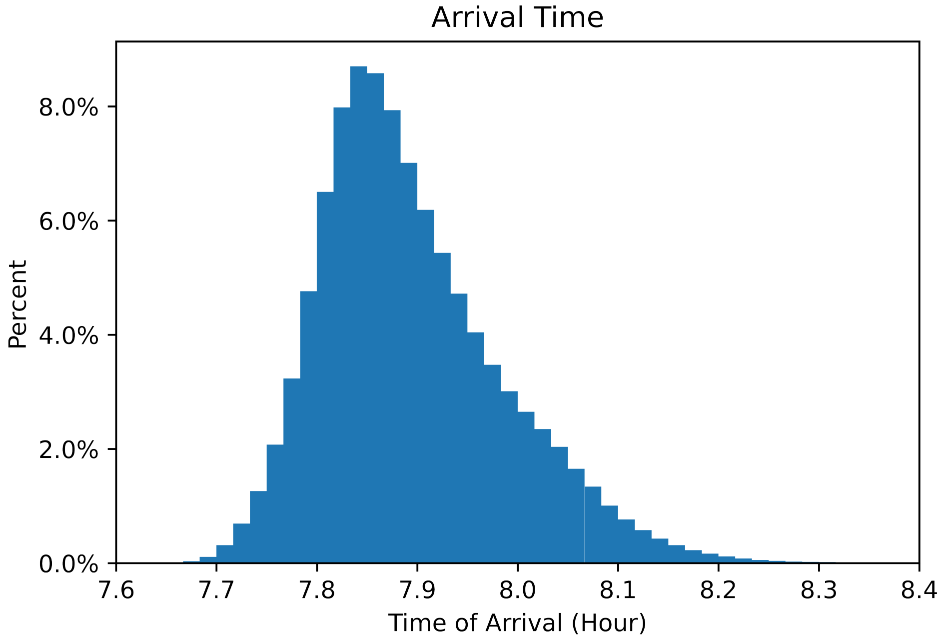

Figure 5

Figure 5 is the results from running the model a million times. Each bar is in a minute segment. As this graph shows, the number of arrivals rapidly increases from 7.7 hours or 7:42 to a peak of around 5% at 7:50. Then it gradually falls from the peak, slowly decreasing. This is somewhatjk similar to a log-normal distribution, however, is distorted. School begins at 8:00, which means any trials greater than 8:00, would be considered late. When dividing the number of times that arrival was later than 8 over by a million, the number of trials in total, the results are 13.9571%. This means that I should expect to be late to school 14% of the time. An interesting thing is that the mean of the times arriving later than 8 is 8:04, or just 4 minutes late. This seems a little modest from experience, as I expected the average to be higher, with much more variation. The reason that this seems better from experience is that the data used for calculating the variables is not perfect and includes more data which is not representative of the times I would actually end up at school on time. For example, as I travel during rush hour, the data also includes the weekends, which do not have as much traffic as the week does. It does not also account for Boston winters, which will cause more variation in the trains and especially buses, where both can encounter delays as a result of the weather. The school calendar year has a greater weight to the winter season than the full year. A way to account for this would be to increase σ and σ^* by 20%. This may not affect the distributions equally, however, they do show if the variation is increased, which is more representative of my actual trip.

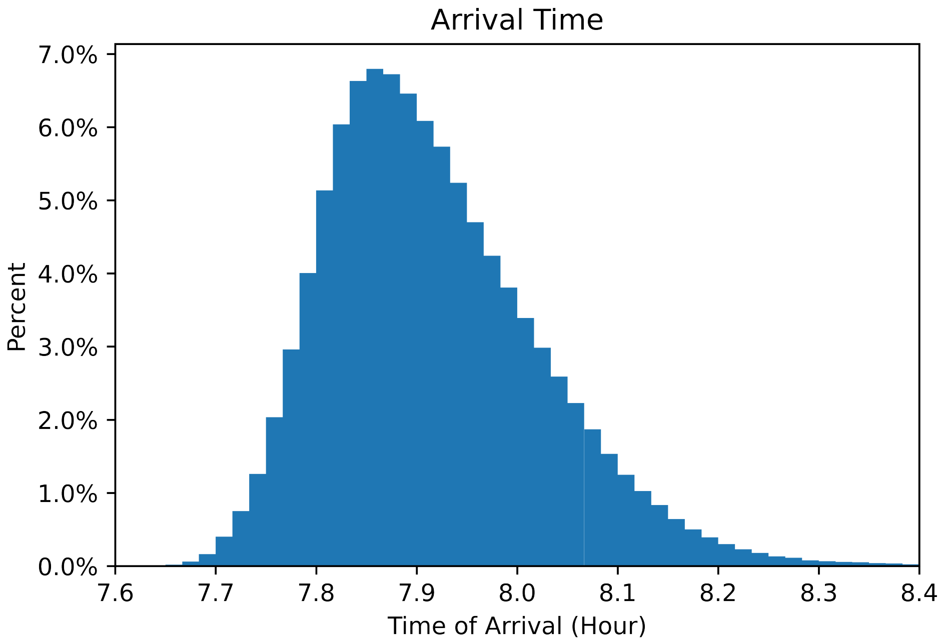

Figure 6

As can be seen in figure 6, this is when the variation is increased, a better model of actual experience. Each bar represents a minute. As can be seen, the peak falls from 5% to 4%, but stays relatively stationary. It is obvious that the tail has increased in size and length. While the mean of the tardy data has increased only from 4 minutes tardy to 5 minutes tardy. The overall tardiness has greatly increased from 14% to 21% (20.749%) of the time. That is a 50% increase, which shows that an increase in the variation greatly increases the number of times being tardy, however, the average does not increase as fast.

5 Conclusion

The results show that I should end up on time to school 86% of the time. In reality this will likely change from season, as during the winter there should be more delays due to snow, higher use, and storms. During the summer it should become more reliable and there should be less interruptions and less intense weather. The weekends are also included in the data, as there was not sufficient time to sort out the weekends. This will also reduce the number of delays, and will shift the data to show it be more reliable. Since this likely composes $\frac{2}{7}$ of the data, it definitely has an effect on the results. Further improvements to the model, would exclude the weekends, as this will not be representative of the trip in interest. By running the model with an increase in the variation to compensate for these effects, as shown in figure 4, found that the model is highly sensitive to the variation. This compensation is not actually derived from any data and was just to see how changes in the variation have on the end result. A better model will also have a better model for determining the waiting time that a person would experience for a train or bus. Both a log-normal and normal distribution were tried and used, however both have flaws and are not the best distributions to represent the data.

References

Massachusetts Bay Transportation Authority. (2019). Mbta travel data [data retrieved from Massachusetts Bay Transportation Authority, https://mbta-massdot.opendata.arcgis.com].

Nicholson, E. (2014, September 21). *The log-normal distribution*. Youtube. https://youtu.be/cLFcqKHT1uY